
(a)
adamw
muon
step
250
500
1000
2000
3000
5000
10000
2.76
2.78
2.75
2.49
d2
d1
d2
d0
1.94
1.76
1.73
d1
d0
1.94
1.95
(b)
JS (bits)
0.15
0.10
0.05
0.00
0
2500
5000
7500
10000
step
(c)
JS (bits)
0.15
0.10
0.05
0.00
4
3
2.5
2
1.75
1.5
1.4
1.3
training loss
AdamW vs Muon
AdamW, other data order
Muon, other data order